The documentation and process conversion measures necessary to comply with this document shall be completed by 15 December 2017.

INCH-POUND

MIL-PRF-19500/500F
15 September 2017
SUPERSEDING
MIL-PRF-19500/500E
w/Amendment 1
12 September 2013

PERFORMANCE SPECIFICATION SHEET

# SEMICONDUCTOR DEVICE, DIODE, SILICON, UNIPOLAR TRANSIENT VOLTAGE SUPPRESSOR, TYPES 1N5555 THROUGH 1N5558, 1N5907, 1N5629 THROUGH 1N5665, JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

## 1. SCOPE

1.1 Scope. This specification covers the performance requirements for 1,500 watt, silicon, transient voltage suppressor diodes. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (DO-13).

1.3 Maximum ratings. Maximum ratings are as shown in maximum test ratings (see 3.6 herein) and as follows:

$P_{PP}$ = 1500 W (see figures 2 and 3) at $t_p$ = 1.0 ms.

$P_{M(AV)}$ = 1.0 W (derate at 6.67 mW/°C above $T_A$ = +25°C) (see 6.5 herein).

$I_{FSM}$ = 200 A (pk) at $t_p$ = 8.3 ms ($T_A$ = +25°C).

-55°C ≤ $T_J$ ≤ +175°C (ambient), -55°C ≤ $T_{STG}$ ≤ +175°C (ambient).

1.4 Primary electrical characteristics at $T_A$ = +25°C. Primary ratings are as shown in maximum test ratings (see 3.6).

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil . Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.



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* 1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.6 for PIN construction example and 6.7 for a list of available PINs.

* 1.5.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: “JAN”, “JANTX”, and "JANTXV".

* 1.5.2 Device type. The designation system for the device types of semiconductor covered by this specification sheet are as follows.

* 1.5.2.1 First number and first letter symbols. The semiconductors of this specification sheet use the first number and letter symbols "1N".

* 1.5.2.2 Second number symbols. The second number symbols for the semiconductor covered by this specification sheet are as follows: "5555" through “5558”, “5907”, and “5629” through “5665”.

* 1.5.3 Suffix symbols. The suffix symbol "A" indicates that the semiconductor is a modified version of the device type.

* 1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QML-19500.

## 2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

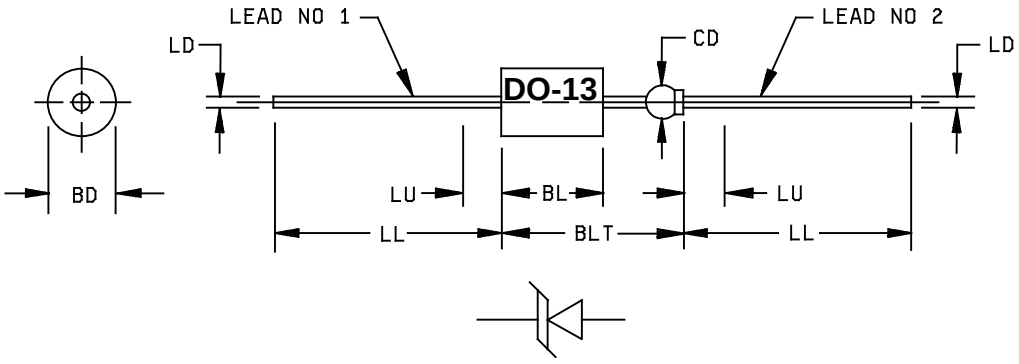
MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://quicksearch.dla.mil.

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



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| Symbol | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| BD | .215 | .235 | 5.46 | 5.97 | |
| BL | .293 | .357 | 7.44 | 9.07 | 3 |
| BLT | | .570 | | 14.48 | |
| CD | .045 | .100 | 1.14 | 2.54 | 5 |
| LD | .025 | .035 | 0.64 | 0.89 | |
| LL | 1.000 | 1.625 | 25.40 | 41.28 | 4 |
| LU | | .188 | | 4.78 | 4 |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. The major diameter is essentially constant along its length.
4. Within this zone, diameter may vary to allow for lead finishes and irregularities.
5. Dimension to allow for pinch or seal deformation anywhere along tubulation.
6. Lead 1 (cathode) shall be electrically connected to the case.
7. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

FIGURE 1. Physical dimensions (DO-13).



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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows: $I_{(BR)}$ - reverse breakdown current at a specified condition.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1 (DO-13) herein.

3.4.1 Metallurgically bonded construction. Metallurgically bonded construction is required. The bonding flow shall have flow points above 260°C.

3.4.2 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Marking. Devices shall be marked in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in 1.3, 1.4, and table I. Maximum ratings shall be in accordance with columns 6 through 8 of table II. Primary electrical characteristics are shown in columns 4 and 5 of table II herein.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I, subgroup 2.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4, and table I, table II, table III, and table IV).

4.1.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-PRF-19500, and as specified herein except, lot accumulation shall be 6-months in lieu of 6-weeks.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table IV tests, the tests specified in table IV herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.



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* 4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen | Measurement |
|---|---|
| | JANTX and JANTXV levels |
| 3a | T(high) = +175°C |
| 9, 10, 11 | Not applicable |
| 12 | See 4.3.1 |
| 13 | Interim electrical, delta, and table I, subgroups 2 and 3, electrical parameters not applicable for this screen (performed in screen 12). |
| 14 | Required |
| 15 | For JANTXV devices only, 100 percent inspection will be in accordance with manufacturer's internal failure criteria. |

* 4.3.1 Power burn-in (HTRB) and steady-state operation life test conditions. The test conditions and order of events shall be as follows:

* a. Pulse in accordance with 4.5.2.a herein, 20 times (screening and group B) and 100 times (group C and E) at $T_A$ = +25°C.

b. Read $I_D$ and $V_{(BR)}$ at $T_A$ = +25°C, remove defective devices and record the number of failures.

c. Apply the working peak reverse voltage ($V_{WM}$) (column 4 of table II herein) at $T_A$ = +125°C as follows:

(1) For 96 hours (JANTX and JANTXV).

(2) For 340 hours (JANTX and JANTXV) for group B steady-state operation life test.

(3) For 1,000 hours for group C steady-state operation life test.

d. Read $I_D$ and $V_{(BR)}$ at $T_A$ = +25°C. Devices with $\Delta I_D$ > 50 percent (100 percent for steady-state operation life) of the initial reading or 1 μA dc, whichever is greater, or $\Delta V_{(BR)}$ > ±2 percent (±5 percent for steady-state operation life) of initial value shall be considered defective. Remove defective devices and record the number of failures.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-195000 and table I herein. End-point electrical measurements shall be in accordance with table I, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500. Electrical measurements (end-points) and Delta requirements shall be in accordance with the applicable steps of table III herein.



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* 4.4.2.1 Group B inspection, table E-VIb of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| B2 | 1051 | -55°C to +175°C. |
| B2 | 4066 | 10 pulses, see 4.5.2 herein. |
| * B3 | 1027 | See 4.3.1, 4.5.1, and 4.5.3. |
| B5 | | Not applicable. |

* 4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500. Electrical measurements (end-points) and Delta requirements shall be in accordance with the applicable steps of table III herein.

4.4.3.1 Group C inspection, table E-VII of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| C2 | 2036 | Lead tension: Test condition A; weight = 10 pounds (4.54 kg); t = 15 ±3 s.<br>Lead fatigue: Test condition E; weight = 8 ounces (226.7 g). |
| C5 | | Not applicable |
| C6 | 1026 | See 4.5.1 and 4.5.2 herein. |
| C7 | 1018 | n = 3, c = 0 or n = 5, c = 1. |
| C8 | | Condition for temperature coefficient of breakdown voltage is as follows:<br>$I_{BR}$ = column 3 of table II herein, $T_1$ = +25°C ±3°C, $T_2$ = $T_1$ +100°C; n = 22, c = 0. |
| C9 | | Condition for maximum peak pulse current is as follows: See 4.5.2.b herein,<br>(20 μs pulse only) 10 pulses; n = 22, c = 0. |

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-IX of MIL-PRF-19500 and as specified herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Group C steady-state operation life test (alternate procedure). When the group B, 340-hour life test is continued on test to 1,000 hours to satisfy the group C life test requirements, the test shall be performed as given in 4.3.1 herein with the following exception: Step 4.3.1.d shall be repeated at the end of the 1,000 hours.



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4.5.2 Maximum peak pulse current ($I_{PP}$). The peak pulse currents specified in column 7 of table II herein shall be applied while simultaneously maintaining a bias voltage of not less than the applicable voltage specified in column 4 of table II, in the same polarity as the peak pulse current. The clamping voltage (Vc) shall be as specified in 4.5.3. The peak pulse current shall be applied with a current vs time waveform as follows (1 pulse per minute maximum):

a. Pulse current shall reach 100 percent of $I_{PP}$ at $t \leq 10$ μs and decay to 50 percent of $I_{PP}$ at $t \geq 1$ ms for $t_p = 1$ ms (see figure 4).

b. Pulse current shall reach 100 percent of $I_{PP}$ at $t \leq 8$ μs and decay to 50 percent of $I_{PP}$ at $t \geq 20$ μs for $t_p = 20$ μs (see figure 5).

4.5.3 Clamping voltage. The peak pulse clamping voltage shall be measured across the diode in a 1 ms time interval. The response detector shall demonstrate equipment accuracy of ±3 percent. The peak clamping voltage as specified in column 6 of table II shall be applicable to the 1 ms pulse of 4.5.2.a only.



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits 2/ | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical examination | 2071 | | | | | |
| Radiography | 2076 | For JANTX devices only. Inspection will be in accordance with manufacturer's internal failure criteria, c = 45, n = 0 | | | | |
| Subgroup 2 | | | | | | |
| Standby current | 4016 | DC method, $V_R = V_{WM}$ (column 4 of table II) | $I_D$ | | Column 5 | μA dc |
| Breakdown voltage | 4022 | $t_p \leq 300$ ms, duty cycle ≤ 2 percent, $I_{BR}$ = column 3 of table II | $V_{BR}$ | Column 2 | Column 2 | V dc |
| Subgroup 3 | | | | | | |
| Low temperature operation: | | $T_A$ = -55°C | | | | |
| Minimum breakdown voltage | 4022 | $t_p \leq 300$ ms, duty cycle ≤ 2 percent, $I_{BR}$ = column 3 of table II | $V_{BR}$ | Column 10 | | V dc |
| High temperature operation: | | TA = 125°C | | | | |
| Reverse current leakage | 4016 | DC method, VR = $V_{WM}$ (column 4 of table II) | $I_{D2}$ | | Column 9 | μA dc |
| Subgroup 4 | | | | | | |
| Clamping voltage maximum | | $t_p$ = 1.0 ms (see 4.5.2.a), $I_{PP}$ = column 7 of table II, (pulsed) (see 4.5.3) | $V_C$ | | Column 6 | V (pk) |
| * Forward voltage | 4011 | Condition B, $I_{FM}$ = 100 A (pk); $t_p$ = 8.3 ms (max), duty cycle = 4 pulses per minute (max) | $V_{FM}$ | | 3.5 | V (pk) |
| Subgroup 5 | | | | | | |
| Not applicable | | | | | | |

See footnotes at end of table.



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* TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits 2/ | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 6 | | | | | | |
| * Forward surge current | 4066 | Condition A, $I_{FSM}$ = 200 A (pk), one pulse, half sine wave (8.3 ms), $I_F$ = 0; $V_{WM}$ = 0; $T_A$ = 25°C | | | | |
| End-point electrical measurements | | See table III, steps 1 and 2 | | | | |
| Subgroup 7 | | | | | | |
| Not applicable | | | | | | |

1/ For sampling plan, see MIL-PRF-19500.
2/ Column references are to table II.



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TABLE II. Characteristics and ratings.

| Col 1 | Col 2 | | Col 3 | Col 4 | Col 5 | Col 6 | Col 7 | | Col 8 | Col 9 | Col 10 | Col 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Type | Breakdown voltage $V_{BR}$ at $I_{BR}$ | | Test current $I_{BR}$ | Working peak reverse voltage $V_{WM}$ | Maximum standby current $I_D$ | Maximum clamping voltage $V_C$ at $I_{PP}$ | Maximum peak pulse current $I_{PP}$ $t_p$ = 20 µs $t_r$ = 8 µs | Maximum peak pulse current $I_{PP}$ $t_p$ = 1 ms $t_r$ = 10 µs | Maximum temperature Coefficient of $V_{BR}$ $\alpha_{VBR}$ | Maximum standby current $I_D$ $T_A$ = +125°C | Minimum Breakdown voltage at $I_{BR}$ $T_A$ = -55°C | Breakdown current $I_{BR}$ maximum dc current $T_A$ = +25°C |
| | V dc Min | V dc Max | mA dc | V (pk) | µA dc | V (pk) | A (pk) | A (pk) | %/°C | µA dc | V dc | mA dc |
| 1N5907 | 6.00 | 6.75 | 1 | 5.00 | 300 | 10.0 | 1,000 | 150.0 | .057 | 1,500 | 5.63 | 140.0 |
| 1N5629A | 6.45 | 7.14 | 10 | 5.80 | 1,000 | 10.5 | 810 | 143.0 | .057 | 4,000 | 6.05 | 140.0 |
| 1N5630A | 7.13 | 7.88 | 10 | 6.40 | 500 | 11.3 | 750 | 132.0 | .061 | 2,000 | 6.66 | 125.0 |
| 1N5631A | 7.79 | 8.61 | 10 | 7.02 | 200 | 12.1 | 700 | 124.0 | .065 | 1,000 | 7.24 | 115.0 |
| 1N5632A | 8.65 | 9.55 | 1 | 7.78 | 50 | 13.4 | 630 | 112.0 | .068 | 400 | 8.01 | 104.0 |
| 1N5633A | 9.50 | 10.50 | 1 | 8.55 | 10 | 14.5 | 585 | 103.5 | .073 | 200 | 8.75 | 95.0 |
| 1N5634A | 10.50 | 11.60 | 1 | 9.40 | 5 | 15.6 | 545 | 96.0 | .075 | 200 | 9.65 | 86.0 |
| 1N5635A | 11.40 | 12.60 | 1 | 10.20 | 5 | 16.7 | 510 | 90.0 | .078 | 200 | 10.40 | 73.0 |
| 1N5636A | 12.40 | 13.70 | 1 | 11.10 | 5 | 18.2 | 465 | 82.0 | .081 | 200 | 11.30 | 70.0 |
| 1N5637A | 14.30 | 15.80 | 1 | 12.80 | 5 | 21.2 | 400 | 71.0 | .084 | 200 | 13.00 | 63.0 |
| 1N5638A | 15.20 | 16.80 | 1 | 13.60 | 5 | 22.5 | 375 | 67.0 | .086 | 200 | 13.70 | 59.0 |
| 1N5639A | 17.10 | 18.90 | 1 | 15.30 | 5 | 25.2 | 335 | 59.5 | .088 | 200 | 15.40 | 53.0 |
| 1N5640A | 19.00 | 21.00 | 1 | 17.10 | 5 | 27.7 | 305 | 54.0 | .090 | 200 | 17.10 | 47.0 |
| 1N5641A | 20.90 | 23.10 | 1 | 18.80 | 5 | 30.6 | 275 | 49.0 | .092 | 200 | 18.80 | 43.0 |
| 1N5642A | 22.80 | 25.20 | 1 | 20.50 | 5 | 33.2 | 255 | 45.0 | .094 | 200 | 20.50 | 39.0 |
| 1N5643A | 25.70 | 28.40 | 1 | 23.10 | 5 | 37.5 | 225 | 40.0 | .096 | 200 | 23.00 | 35.0 |
| 1N5644A | 28.50 | 31.50 | 1 | 25.60 | 5 | 41.4 | 205 | 36.0 | .097 | 200 | 23.50 | 31.0 |
| 1N5645A | 31.40 | 34.70 | 1 | 28.20 | 5 | 45.7 | 185 | 33.0 | .098 | 200 | 28.00 | 28.0 |
| 1N5555 | 33.00 | | 1 | 30.50 | 5 | 47.5 | 193 | 32.0 | .093 | 200 | 30.20 | 27.0 |
| 1N5646A | 34.20 | 37.80 | 1 | 30.80 | 5 | 49.9 | 170 | 30.0 | .099 | 200 | 30.50 | 26.0 |





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TABLE II. Characteristics and ratings - Continued.

| Col 1 | Col 2 | | Col 3 | Col 4 | Col 5 | Col 6 | Col 7 | | Col 8 | Col 9 | Col 10 | Col 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Type | Breakdown Voltage $V_{BR}$ at $I_{BR}$ | | Test current $I_{BR}$ | Working peak reverse voltage $V_{WM}$ | Maximum standby current $I_D$ | Maximum clamping voltage $V_C$ at $I_{PP}$ | Maximum peak pulse current $I_{PP}$ $t_p$ = 20 µs $t_r$ = 8 µs | Maximum peak pulse current $I_{PP}$ $t_p$ = 1 ms $t_r$ = 10 µs | Maximum temperature coefficient of $V_{BR}$ $\alpha V_{BR}$ | Maximum standby current $I_D$ $T_A$ = +125°C | Minimum breakdown voltage at $I_{BR}$ $T_A$ = -55°C | Breakdown current $I_{BR}$ maximum dc current $T_A$ = +25°C |
| | V dc Min | V dc Max | mA dc | V (pk) | µA dc | V (pk) | A (pk) | A (pk) | %/°C | µA dc | V dc | mA dc |
| 1N5647A | 37.10 | 41.00 | 1 | 33.30 | 5 | 53.9 | 155 | 28.0 | .100 | 200 | 33.10 | 24.0 |
| 1N5648A | 40.90 | 45.20 | 1 | 36.80 | 5 | 59.3 | 145 | 25.3 | .101 | 200 | 36.40 | 22.0 |
| 1N5556 | 43.70 | | 1 | 40.30 | 5 | 63.5 | 136 | 24.0 | .094 | 200 | 40.00 | 21.0 |
| 1N5649A | 44.70 | 49.40 | 1 | 40.20 | 5 | 64.8 | 130 | 23.2 | .101 | 200 | 39.80 | 20.0 |
| 1N5650A | 48.50 | 53.60 | 1 | 43.60 | 5 | 70.1 | 120 | 21.4 | .102 | 200 | 43.10 | 18.0 |
| 1N5651A | 53.20 | 58.80 | 1 | 47.80 | 5 | 77.0 | 110 | 19.5 | .103 | 200 | 47.30 | 17.0 |
| 1N5557 | 54.00 | | 1 | 49.00 | 5 | 78.5 | 116 | 19.0 | .096 | 200 | 48.50 | 16.0 |
| 1N5652A | 58.90 | 65.10 | 1 | 53.00 | 5 | 85.0 | 100 | 17.7 | .104 | 200 | 52.30 | 15.0 |
| 1N5653A | 64.60 | 71.40 | 1 | 58.10 | 5 | 92.0 | 90 | 16.3 | .104 | 200 | 57.30 | 14.0 |
| 1N5654A | 71.30 | 78.80 | 1 | 64.10 | 5 | 103.0 | 82 | 14.6 | .105 | 200 | 63.20 | 12.0 |
| 1N5655A | 77.90 | 86.10 | 1 | 70.10 | 5 | 113.0 | 75 | 13.3 | .105 | 200 | 69.00 | 11.0 |
| 1N5656A | 86.50 | 95.50 | 1 | 77.80 | 5 | 125.0 | 68 | 12.0 | .106 | 200 | 76.50 | 10.0 |
| 1N5657A | 95.00 | 105.00 | 1 | 85.50 | 5 | 137.0 | 62 | 11.0 | .106 | 200 | 84.10 | 9.5 |
| 1N5658A | 105.00 | 116.00 | 1 | 94.00 | 5 | 152.0 | 55 | 9.9 | .107 | 200 | 92.80 | 8.5 |
| 1N5659A | 114.00 | 126.00 | 1 | 102.00 | 5 | 165.0 | 50 | 9.1 | .107 | 200 | 100.00 | 7.5 |
| 1N5660A | 124.00 | 137.00 | 1 | 111.00 | 5 | 179.0 | 47 | 8.4 | .107 | 200 | 109.00 | 7.0 |
| 1N5661A | 143.00 | 158.00 | 1 | 128.00 | 5 | 207.0 | 40 | 7.2 | .108 | 200 | 126.00 | 6.0 |
| 1N5662A | 152.00 | 168.00 | 1 | 136.00 | 5 | 219.0 | 38 | 6.8 | .108 | 200 | 134.00 | 5.8 |
| 1N5663A | 162.00 | 179.00 | 1 | 145.00 | 5 | 234.0 | 36 | 6.4 | .108 | 200 | 143.00 | 5.5 |
| 1N5664A | 171.00 | 189.00 | 1 | 154.00 | 5 | 246.0 | 34 | 6.1 | .108 | 200 | 151.00 | 5.0 |
| 1N5665A | 190.00 | 210.00 | 1 | 171.00 | 5 | 274.0 | 30 | 5.5 | .108 | 200 | 167.00 | 4.5 |
| 1N5558 | 191.00 | | 1 | 175.00 | 5 | 265.0 | 33 | 5.7 | .100 | 200 | 172.00 | 4.5 |





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TABLE III. Groups B and C electrical and delta measurements. 1/ 2/ 3/

| Step | Inspection | MIL-STD-750 | | Symbol | Limits 4/ | | Unit |
|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | Min | Max | |
| 1. | Standby current | 4016 | DC method, $V_R = V_{WM}$ (column 4 of table II) | $I_D$ | | Column 5 | μA dc |
| 2. | Breakdown voltage | 4022 | $t_p \leq 300$ ms, duty cycle ≤ 2 percent, $I_{BR}$ = column 3 of table II | $V_{BR}$ | Column 2 | Column 2 | V dc |
| 3. | Standby current | 4016 | DC method; $V_R = V_{WM}$ (column 4 of table II) | $\Delta I_D$ | | 100 percent of initial reading or 20 percent of column 5 of table II, whichever is greater. | |
| 4. | Breakdown voltage | 4022 | $t_p \leq 300$ ms, duty cycle ≤ 2 percent, $I_{BR}$ = column 3 of table II | $\Delta V_{BR}$ | | ± 5 percent of initial value | |
| 5. | Clamping voltage | | $t_p$ = 1.0 ms (see 4.5.2.a); $I_{PP}$ = column 7 of table II | $V_C$ | | Column 6 | V (pk) |

1/ The electrical measurements for table E-VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 are as follows:
   a. Subgroup 2, see table III herein, steps 1 and 2.
   b. Subgroup 6, see table III herein, steps 1, 2, 3, and 4.

2/ The electrical measurements for table E-VII (JAN, JANTX, and JANTXV) of MIL-PRF-19500 are as follows:
   a. Subgroup 2 and 3, see table III herein, and steps 1 and 2.
   b. Subgroup 9, see table III herein, steps 1 and 2 for all levels.

3/ Column references are to table II.

4/ Devices which exceed the table I limits for this test shall not be accepted.



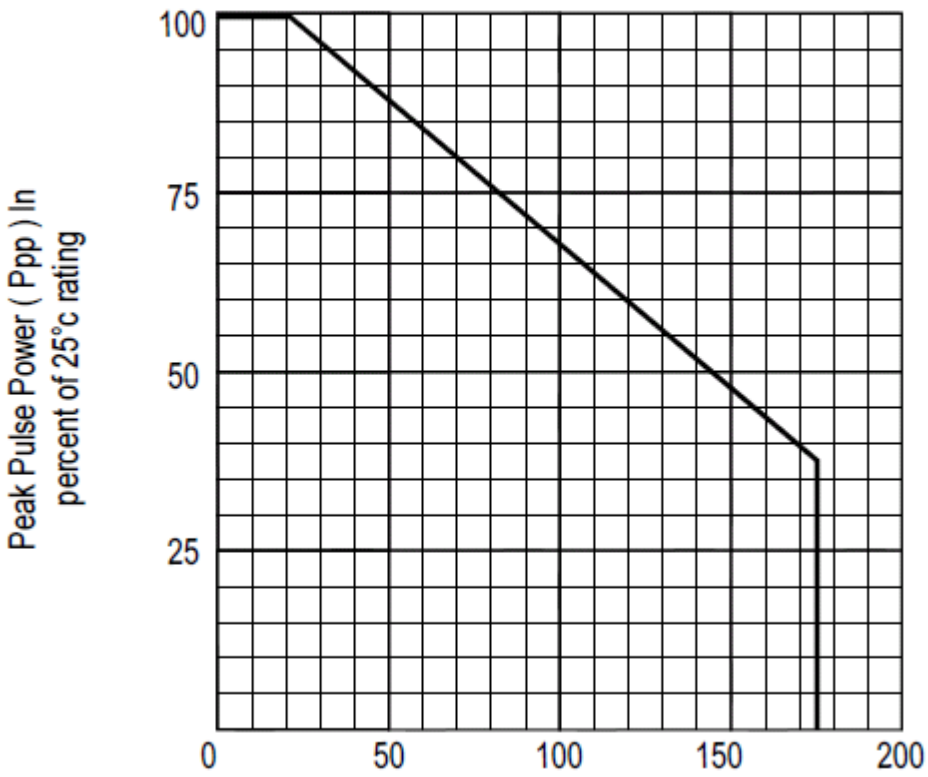
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TABLE IV. Group E inspection (all quality levels) for qualification only.

| Inspection | MIL-STD-750 | | Sampling plan |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | n = 45, c = 0 |
| Temperature cycling | 1051 | 500 cycles, condition C, -55°C to +175°C. | |
| Electrical measurements | | See table III, steps 1, 2, 3, and 4. | |
| Subgroup 2 | | | n = 22, c = 0 |
| Life test | | 1,000 hours. See 4.3.1 | |
| Electrical measurements | | See table III, steps 1, 2, 3, and 4. | |
| Subgroups 4 and 5 | | | |
| Not applicable | | | |
| Subgroup 6 | | | |
| Peak pulse current | | See 4.5.2. $I_{PP}$ shall be characterized by the supplier and this data shall be available to the government. Test shall be performed on each low and high voltage device for each structurally identical grouping. Test to failure. | |
| Electrical measurements | | See table III, steps 1, 2, 3, and 4. | |
| Subgroup 7 | | | n = 45, c = 0 |
| Soldering heat | 2031 | 1 cycle. | |



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* FIGURE 2. Derating curve.



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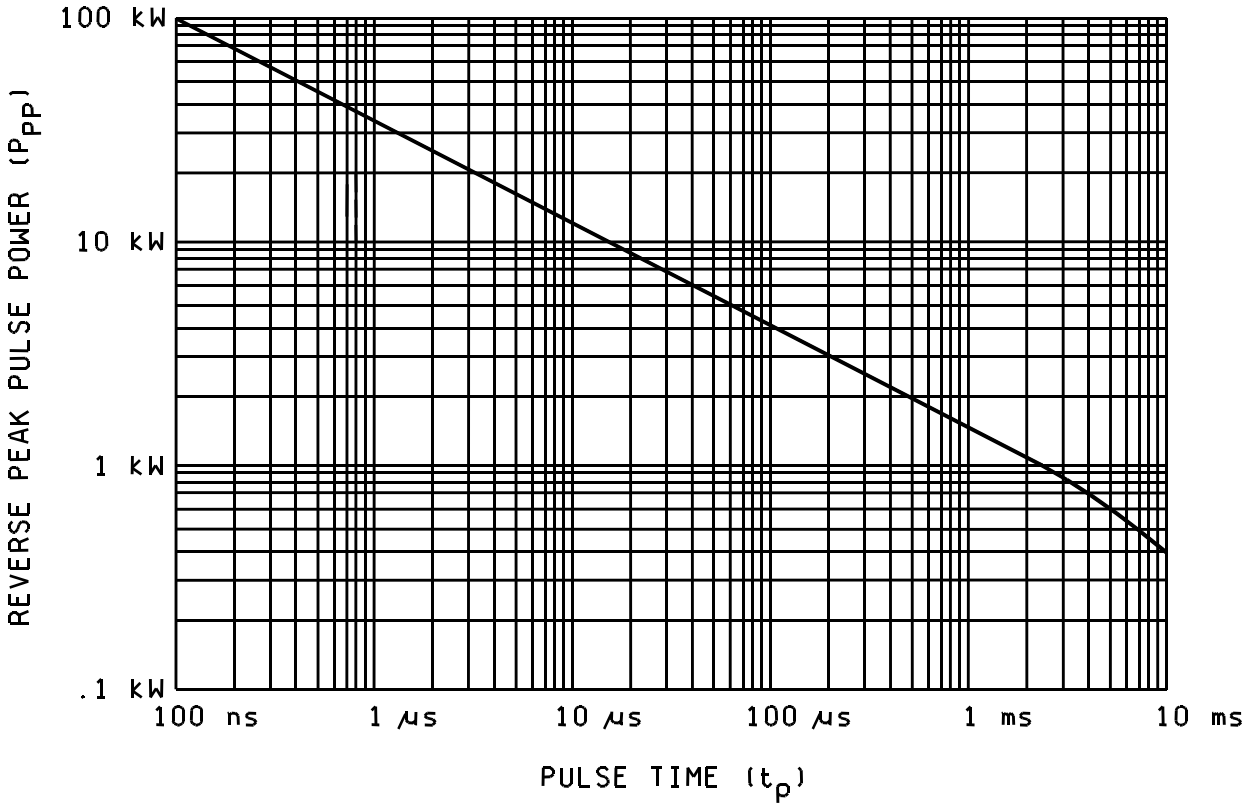
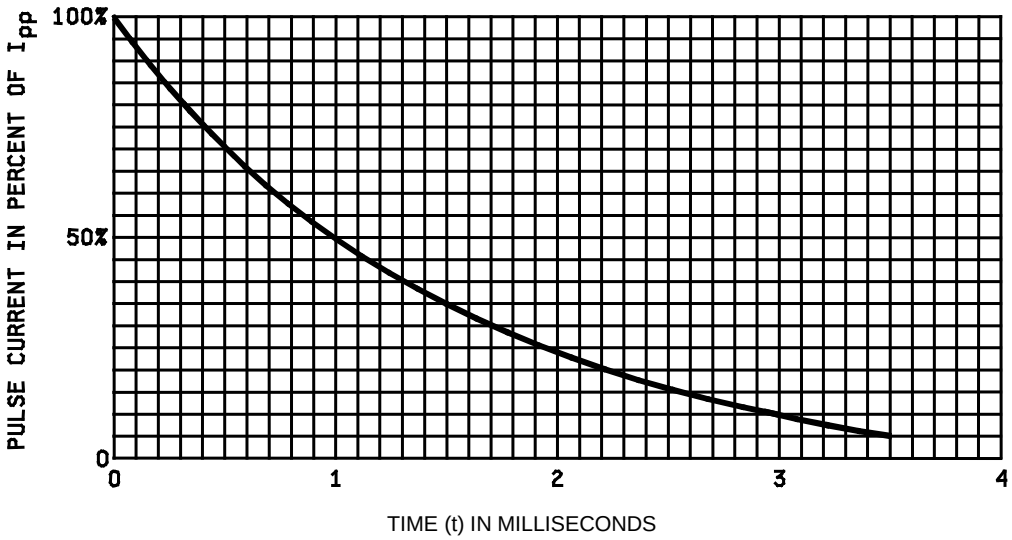



FIGURE 3. Peak pulse power versus pulse time.

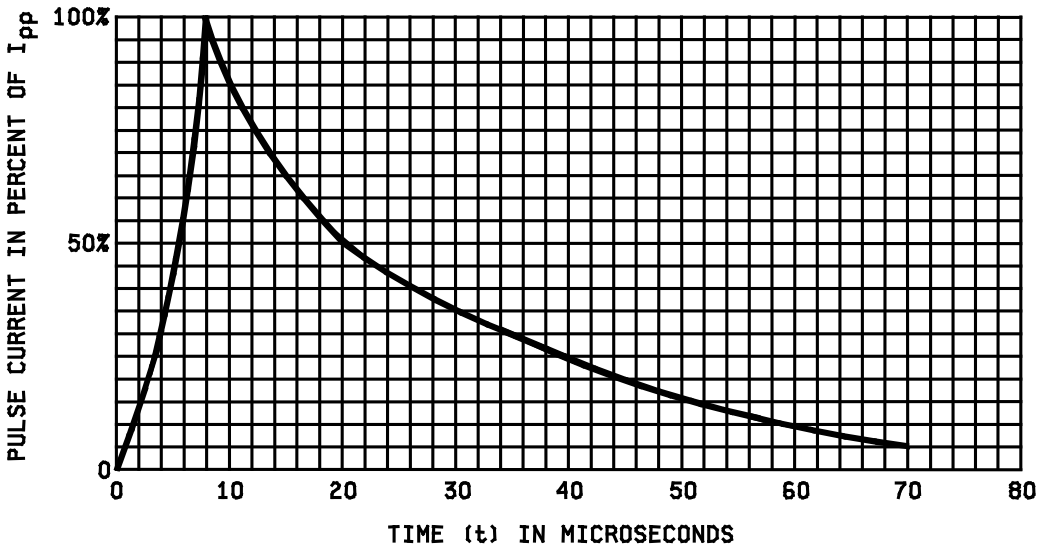


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NOTE: Pulse time duration is defined as that point where the pulse current decays to 50 percent of $I_{PP}$. (Rise time to 100 percent of $I_{PP}$ = 10 μs).

FIGURE 4. Current impulse waveform (see 4.5.2.a herein).




NOTE: Pulse time duration is defined as that point where the pulse current decays to 50 percent of $I_{PP}$. (Rise time to 100 percent of $I_{PP}$ = 8 μs).

FIGURE 5. Current impulse waveform (see 4.5.2.b herein).



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.2).

* d. The complete Part or Identifying Number (PIN), see title and section 1.

e. Destructive physical analysis when requested.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Substitution information. Type numbers 1N5555 through 1N5558 were previously covered by MIL-PRF-19500/434.

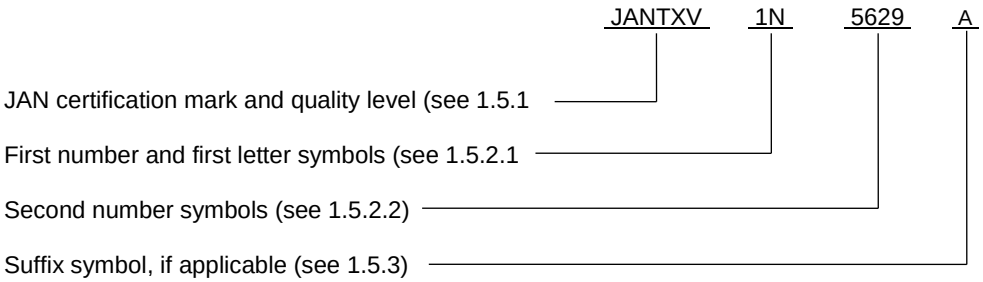
6.5 Steady state power rating. This rating is not relevant for most applications.



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* 6.6 PIN construction example.

* 6.6.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.






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* 6.7 List of PINs. The following is a list of possible PINs available on this specification sheet.

| PINs for types 1N5555 through 1N5558, 1N5907, 1N5629A THROUGH 1N5665A. | | |
|---|---|---|
| JAN1N5555 | JANTX1N5555 | JANTXV1N5555 |
| JAN1N5556 | JANTX1N5556 | JANTXV1N5556 |
| JAN1N5557 | JANTX1N5557 | JANTXV1N5557 |
| JAN1N5558 | JANTX1N5558 | JANTXV1N5558 |
| JAN1N5907 | JANTX1N5907 | JANTXV1N5907 |
| JAN1N5629A | JANTX1N5629A | JANTXV1N5629A |
| JAN1N5630A | JANTX1N5630A | JANTXV1N5630A |
| JAN1N5631A | JANTX1N5631A | JANTXV1N5631A |
| JAN1N5632A | JANTX1N5632A | JANTXV1N5632A |
| JAN1N5633A | JANTX1N5633A | JANTXV1N5633A |
| JAN1N5634A | JANTX1N5634A | JANTXV1N5634A |
| JAN1N5635A | JANTX1N5635A | JANTXV1N5635A |
| JAN1N5636A | JANTX1N5636A | JANTXV1N5636A |
| JAN1N5637A | JANTX1N5637A | JANTXV1N5637A |
| JAN1N5638A | JANTX1N5638A | JANTXV1N5638A |
| JAN1N5639A | JANTX1N5639A | JANTXV1N5639A |
| JAN1N5640A | JANTX1N5640A | JANTXV1N5640A |
| JAN1N5641A | JANTX1N5641A | JANTXV1N5641A |
| JAN1N5642A | JANTX1N5642A | JANTXV1N5642A |
| JAN1N5643A | JANTX1N5643A | JANTXV1N5643A |
| JAN1N5644A | JANTX1N5644A | JANTXV1N5644A |
| JAN1N5645A | JANTX1N5645A | JANTXV1N5645A |
| JAN1N5646A | JANTX1N5646A | JANTXV1N5646A |
| JAN1N5647A | JANTX1N5647A | JANTXV1N5647A |
| JAN1N5648A | JANTX1N5648A | JANTXV1N5648A |
| JAN1N5649A | JANTX1N5649A | JANTXV1N5649A |
| JAN1N5650A | JANTX1N5650A | JANTXV1N5650A |
| JAN1N5651A | JANTX1N5651A | JANTXV1N5651A |
| JAN1N5652A | JANTX1N5652A | JANTXV1N5652A |
| JAN1N5653A | JANTX1N5653A | JANTXV1N5653A |
| JAN1N5654A | JANTX1N5654A | JANTXV1N5654A |
| JAN1N5655A | JANTX1N5655A | JANTXV1N5655A |
| JAN1N5656A | JANTX1N5656A | JANTXV1N5656A |
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| JAN1N5660A | JANTX1N5660A | JANTXV1N5660A |
| JAN1N5661A | JANTX1N5661A | JANTXV1N5661A |
| JAN1N5662A | JANTX1N5662A | JANTXV1N5662A |
| JAN1N5663A | JANTX1N5663A | JANTXV1N5663A |
| JAN1N5664A | JANTX1N5664A | JANTXV1N5664A |
| JAN1N5665A | JANTX1N5665A | JANTXV1N5665A |



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* 6.8 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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